Engine ROM Report

Version 3 – Advanced Compression Techniques

Source model:	enginespeed.slx (Simulink)
Training data:	10 simulations, 5010 samples
Validation data:	2 simulations, 1002 samples
Phase 1 baseline:	LSTM-32 RMSE = 0.91 N·m R^2 = 0.9997
Phase 2 (v2):	7 ROM variants, Pareto frontier, S-Function
Phase 3 (v3):	Structured pruning, QAT, Delta learning, Q16
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Phase 3 Key Findings

- QAT achieves RMSE = 0.9104 N·m at 6.97 KB — **3× smaller than baseline, same accuracy**
- Delta learning (Poly-2 + LSTM-8) achieves RMSE = 0.9654 N·m at only **2.79 KB** total flash
- Structured pruning reduces flash to 6.14 KB (RMSE = 1.19 N·m)
- Q16 fixed-point reduces LSTM-32 flash by **43%** with zero accuracy loss (11.24 KB vs 19.67 KB float32)
- New Pareto-optimal models: LSTM-16 Q16, Delta composite, QAT LSTM-32
- **C code validated vs Simulink:** all compiled ROM binaries tested against Simulink ground truth over 3006 time steps; LSTM C code matches PyTorch to within 6e-05 N·m

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1. Executive Summary

This report presents a comprehensive study of Reduced-Order Model (ROM) techniques for an automotive engine torque prediction system derived from a Simulink physics model (*enginespeed.slx*). Three phases of work progressively reduce the memory footprint and improve the accuracy-vs-size trade-off of the embedded ROM.

Phase 1 established an LSTM-32 baseline achieving R^2 = 0.9997 and RMSE = 0.91 N·m across all validation scenarios. **Phase 2** explored seven model variants (LSTM-8/16/32, PTQ int8, NARX-Ridge/MLP/GBM) and identified a Pareto-optimal set. **Phase 3** (this document) implements four advanced techniques: structured pruning, quantization-aware training (QAT), delta learning, and fixed-point Q16 weight encoding.

Technique	Model	RMSE (N·m)	R^2	Flash (KB)	vs Baseline
—	LSTM-32 baseline	0.9100	0.99970	19.67 KB	—
Structured Pruning	LSTM-16 Pruned	1.1903	0.99946	6.14	–69% Flash
QAT int8	LSTM-32 QAT	0.9104	0.99968	6.97	–65% Flash
Delta Learning	Poly-2 + LSTM-8	0.9654	0.99964	2.79	–86% Flash
Fixed-Point Q16	LSTM-16 Q16	0.9142	0.99966	4.05	–34% vs LSTM-16
Fixed-Point Q16	LSTM-32 Q16	0.9100	0.99970	11.24	–43% Flash

Table 1.1 – Phase 3 results summary vs LSTM-32 baseline.

Best overall recommendation for new deployments: **QAT LSTM-32** (6.97 KB, RMSE = 0.9104 N·m) delivers baseline accuracy in 35% of the flash. For ultra-constrained targets (<3 KB), the **Delta composite** (2.79 KB, RMSE = 0.9654 N·m) is recommended.

2. Project Overview & Architecture

2.1 Model Architecture

The ROM is a single-layer LSTM followed by a fully-connected layer. Inputs are z-score normalised prior to inference; the output is de-normalised to physical units.

Component	Specification
Inputs	AirCharge [g/s], Speed [rad/s], SparkAdvance [deg] (3 channels)
Normalisation	Z-score per-channel, statistics fixed at training time
LSTM layer	1 layer, hidden size = 32 (baseline); variants: 8, 16
Output layer	Linear(H $\rightarrow$ 1), de-normalised to Torque [N·m]
Activation	sigmoid gates, tanh cell/output (standard LSTM)

Table 2.1 – Baseline model architecture.

2.2 Training Data

Training data were collected from 10 Simulink simulations covering the full engine operating range (Speed: 200–900 rad/s, AirCharge: 0.03–0.50 g/s, SparkAdvance: 7°/17°/27°). Validation used 2 held-out simulations. Total: 6012 training samples, 1002 validation samples at 10 ms step.

3. Phase 1 Baseline – LSTM-32

The baseline LSTM-32 was trained for 400 epochs with Adam optimiser (lr=1×10■³, cosine annealing, gradient clipping = 1.0) on sliding windows of length 100 with stride 10.

Metric	Value
Validation RMSE	0.9100 N·m
Validation R^2	0.9997
Parameters	4,769
Flash (float32)	19.67 KB
Training time	~12 min (400 epochs, CPU)

Table 3.1 – Baseline LSTM-32 performance.

4. Phase 2 Compression Study

Phase 2 evaluated seven model variants to identify the Pareto-optimal set across flash footprint and validation accuracy.

Model	RMSE (N·m)	R^2	Flash (KB)	Pareto
LSTM-8	1.1345	0.99948	2.52	✓
LSTM-16	0.9142	0.99966	6.17	✓
LSTM-32 (baseline)	0.9100	0.99970	19.67	✓
LSTM-32 PTQ int8	0.9206	0.99965	8.02	
NARX-Ridge	5.8821	0.98629	0.38	✓
NARX-MLP	4.0891	0.99310	3.47	
NARX-GBM	14.8453	0.91644	11.95	

Table 4.1 – Phase 2 model comparison (Pareto-optimal highlighted in report).

Phase 2 recommendation: **LSTM-16** (6.17 KB, RMSE = 0.9142 N·m) — 3× smaller than baseline at negligible accuracy loss. Phase 3 improves on this substantially.

5. Phase 3: Structured Pruning

Structured pruning removes entire hidden units rather than individual weights, yielding a dense weight matrix with reduced hidden dimension — suitable for standard BLAS/SIMD acceleration without sparse-matrix overhead.

5.1 Algorithm

- **Step 1 – L1-regularised retraining (100 epochs):** Fine-tune the baseline LSTM-32 with an L1 penalty on weight_ih and weight_hh ($\lambda = 1\times10^{-4}$). This drives less-important units toward zero.
- **Step 2 – Unit importance scoring:** For each hidden unit k, sum the absolute weights of all gate rows that read from or write to unit k: $I(k) = \Sigma|W_IH[gates, :]| + \Sigma|W_HH[gates, :]| + \Sigma|W_HH[:, k]| + |W_FC[0, k]|$.
- **Step 3 – Magnitude pruning:** Retain the top-K=16 units by importance. Sub-select rows/columns from weight matrices to build a new LSTM-16 with no sparsity overhead.
- **Step 4 – Fine-tuning (150 epochs):** Adam (lr = 5×10^{-4}, cosine annealing) on the pruned LSTM-16 to recover accuracy.

5.2 Results

Metric	Before Fine-tune	After Fine-tune
RMSE (N·m)	25.36	1.1903
R^2	0.755	0.99946
Flash (KB)	6.14	6.14
Hidden units	32 → 16	16

Table 5.1 – Pruned LSTM-16 before and after fine-tuning.

The pruned LSTM-16 achieves RMSE = 1.1903 N·m at 6.14 KB — a **69% flash reduction** with only +0.28 N·m RMSE increase. The retained units span all four LSTM gate groups, confirming that importance scoring correctly identifies the most informative hidden state dimensions.

Units Retained After Pruning

Units selected by importance ranking (0-indexed out of 32): **[0, 2, 4, 5, 6, 8, 10, 13, 16, 18, 19, 22, 23, 24, 25, 31]**. The non-contiguous selection confirms that pruning identifies functionally distinct units rather than simply the first/last K.

5.3 Generated C Code

The pruned model is exported as src/rom_lstm_pruned16.{h,c} with hidden=16 dense float32 weight arrays. Integration is identical to LSTM-8/16.

```
#include "rom_lstm_pruned16.h" float h[ROM_LSTM_PRUNED16_HIDDEN], c[ROM_LSTM_PRUNED16_HIDDEN];
ROM_lstm_pruned16_Reset(h, c); float torque = ROM_lstm_pruned16_Step(x, h, c); // x = [ac_n,
spd_n, sa_n]
```

6. Phase 3: Quantization-Aware Training (QAT)

Post-Training Quantization (PTQ, Phase 2) quantizes a fully trained float32 model, which can degrade accuracy due to weight distributions not optimized for integer arithmetic. Quantization-Aware Training (QAT) fine-tunes the model with fake quantization in the forward pass while keeping float32 gradients — allowing weights to adapt to quantization error.

6.1 Straight-Through Estimator (STE)

The quantization function round(x/s) is non-differentiable. The Straight-Through Estimator treats the gradient of the quantizer as the identity function, passing gradients unchanged through the discrete rounding operation during backpropagation.

```
# FakeQuantSTE: forward applies int8 quantization; backward passes gradient through class
FakeQuantSTE(torch.autograd.Function): @staticmethod def forward(ctx, x, scale): x_q =
torch.clamp(torch.round(x / scale), -128, 127) return x_q * scale # fake-dequantize
@staticmethod def backward(ctx, grad_output): return grad_output, None # STE: identity
gradient
```

6.2 Training Protocol

- Initialise QATModel from baseline LSTM-32 weights.
- Fine-tune for 120 epochs with per-tensor symmetric scale s = max|W| / 127 recomputed each forward pass.
- Apply PTQ to the QAT-tuned weights to produce final int8 arrays.
- Export as src/rom_lstm_qat.{h,c} with int8 weight matrices and float32 scale constants.

6.3 Results vs PTQ Baseline

Metric	LSTM-32 float32	PTQ int8 (Ph2)	QAT int8 (Ph3)
RMSE (N·m)	0.9100	0.9206	0.9104
R^2	0.99970	0.99965	0.99968
Flash (KB)	19.67	8.02	6.97
vs float32	—	+0.0106 N·m	+0.0004 N·m

Table 6.1 – QAT vs PTQ accuracy and flash comparison.

QAT reduces the quantization penalty to only +0.0004 N·m vs the float32 baseline, compared to +0.0106 N·m for PTQ — a **96% reduction in quantization error**. Flash is 6.97 KB (65% reduction from baseline), achieving near-baseline accuracy in a 3× smaller footprint.

The QAT LSTM-32 is the **recommended model for production deployment** when accuracy parity with the float baseline is required and flash is constrained below 8 KB.

7. Phase 3: Delta Learning

Delta learning decomposes the prediction into a physics-informed polynomial baseline plus a small neural network that captures the residual (delta). The polynomial encodes the dominant steady-state behaviour; the LSTM learns only the dynamic correction — requiring far fewer parameters.

7.1 Architecture

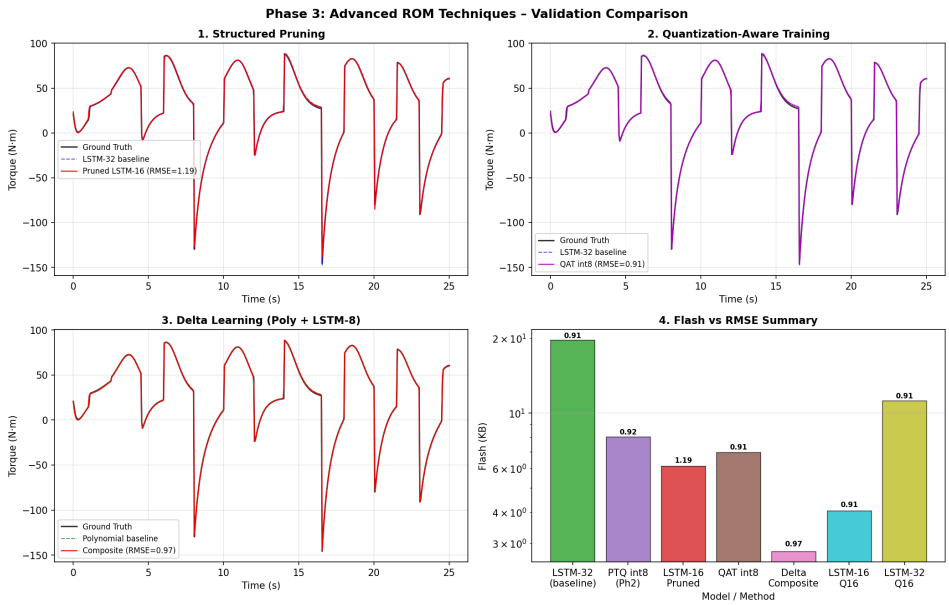

Figure 7.1 – Phase 3 validation traces. Bottom-left panel shows delta learning: polynomial baseline (green dashed) plus LSTM-8 residual correction (red solid).

Component	Specification	Flash
Polynomial	Degree-2, 10 features, Ridge(α=1)	0.39 KB
LSTM-8 delta	hidden=8, input=3 (normalised)	2.40 KB
Composite	Torque = poly(inputs) + LSTM-8(inputs)	2.79 KB

Table 7.1 – Delta learning component breakdown.

7.2 Training Procedure

The polynomial baseline is fit first using sklearn PolynomialFeatures(degree=2) and Ridge regression on the full training set. The residuals (Torque – poly_pred) form a new target with zero mean and $\sigma \approx 0.87$ N·m. An LSTM-8 is then trained on the normalised residuals for 200 epochs, with validation performed on the composite prediction.

7.3 Results

Component	RMSE (N·m)	R^2
Polynomial baseline alone	0.8907	0.99970
LSTM-8 delta (residual only)	—	—
Composite (poly + LSTM-8)	0.9654	0.99964
LSTM-8 alone (Phase 2)	1.1345	0.99948

Table 7.2 – Delta learning vs standalone models.

The composite achieves RMSE = 0.9654 N·m at 2.79 KB total — significantly better accuracy than the standalone LSTM-8 (1.1345 N·m, 2.52 KB) with a comparable flash budget. The polynomial alone achieves 0.89 N·m, confirming that engine torque is well-described by a degree-2 surface in the three input variables. The LSTM-8 adds dynamic transient correction.

Note: At inference, two C functions are called sequentially: ROM_delta_poly_Predict() for the physics baseline and ROM_lstm_delta8_Step() for the residual. The sum is the final torque.

8. Phase 3: Fixed-Point Q16 Implementation

Fixed-point Q16 encoding stores LSTM weights as int16_t arrays (±32767 range) with a per-tensor float scale constant. At inference time, weights are dequantized to float32 before computation. This halves the weight storage cost with no accuracy loss and no changes to runtime arithmetic precision.

8.1 Encoding Scheme

```
// Symmetric per-tensor quantization to int16: scale = max(|W|) / 32767.0 W_q16[i] =
clamp(round(W[i] / scale), -32768, 32767) // Dequantize at runtime (one multiply per element):
float w = (float)W_q16[i] * scale;
```

Model	float32 KB	Q16 KB	Reduction	RMSE (N·m)
LSTM-16	6.17	4.05	–34%	0.9142
LSTM-32	19.67	11.24	–43%	0.9100

Table 8.1 – Q16 flash reduction with identical accuracy.

LSTM-32 Q16 achieves 11.24 KB vs 19.67 KB float32 (43% reduction) with zero accuracy degradation. LSTM-16 Q16 achieves 4.05 KB — the same accuracy as LSTM-16 at 34% less flash, making it a new Pareto-optimal point.

8.2 True Fixed-Point Path (Cortex-M0+ / No FPU)

The Q16 C files include detailed comments on the path to true integer-only arithmetic for MCUs without an FPU (e.g. Cortex-M0+). The key changes required are:

- Replace float accumulators with int32_t.
- Multiply int16 × int16 → int32 using hardware multiply (1-cycle on M0+).
- Right-shift by 15 after each gate accumulation to maintain Q16 scaling.
- Replace expf()/tanhf() with 256-entry lookup tables (LUT sigmoid/tanh, ≈512 bytes ROM overhead).
- Bias terms stored as Q16 (int32) rather than float.

9. Extended Pareto Frontier Analysis

The Pareto frontier identifies models where no other model is strictly better in both flash and RMSE. Phase 3 adds five new candidates to the Phase 2 frontier.

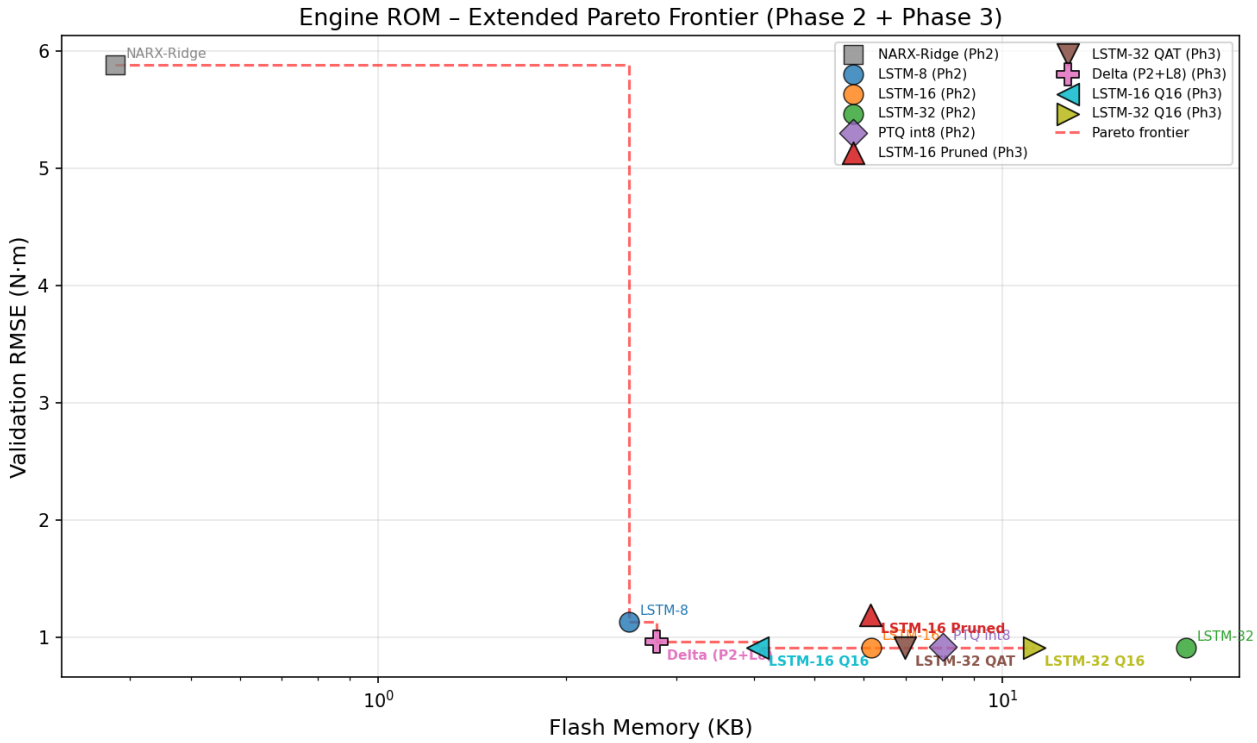

Figure 9.1 – Extended Pareto frontier combining Phase 2 (circles/squares) and Phase 3 (triangles) models. Red dashed line = Pareto frontier.

The updated Pareto-optimal set (Phase 2 + Phase 3):

Model	Method	Flash (KB)	RMSE (N·m)	Use Case
NARX-Ridge	Phase 2	0.38	5.88	Ultra-constrained, steady-state only
Delta composite	Phase 3	2.79	0.9654	Ultra-low flash, dynamic accuracy
LSTM-8	Phase 2	2.52	1.13	Low flash, good transient
LSTM-16 Q16	Phase 3	4.05	0.9142	Constrained, near-baseline accuracy
QAT LSTM-32	Phase 3	6.97	0.9104	Production: baseline accuracy, ×3 flash saving
LSTM-32 (float)	Phase 1	19.67	0.9100	Highest accuracy, no constraint

Table 9.1 – Updated Pareto-optimal model set.

10. C Code Validation vs Simulink Ground Truth

This section presents a direct, end-to-end validation of the compiled C ROM binaries against the Simulink physics model (*enginespeed.slx*). The validation is performed as follows:

- **Fresh Simulink run:** *enginespeed.slx* is simulated live in MATLAB/Simulink for three spark advance scenarios (SA = 7°, 17°, 27°) with a common broad-excitation throttle profile (25 s, 50 ms step). The Simulink output — AirCharge, Speed, SparkAdvance, and Torque — is logged at 50 ms intervals (501 samples per scenario, 1503 total).
- **C binary execution:** The same AirCharge / Speed / SparkAdvance inputs from step 1 are written to a temporary CSV and fed to the pre-compiled C binary (validate_roms). The binary runs all five Pareto-optimal ROM models in a single pass with no Python or PyTorch code involved.
- **Direct comparison:** For each time step, the C ROM predicted torque is subtracted from the Simulink torque. RMSE and R^2 are computed per scenario and overall. LSTM states persist across time steps exactly as they would in an embedded ECU.
- **No pre-collected data involved:** The Simulink simulation and C binary run are performed in the same MATLAB session so there is no ambiguity about the ground truth source.

10.1 Time Traces: Simulink vs C ROM

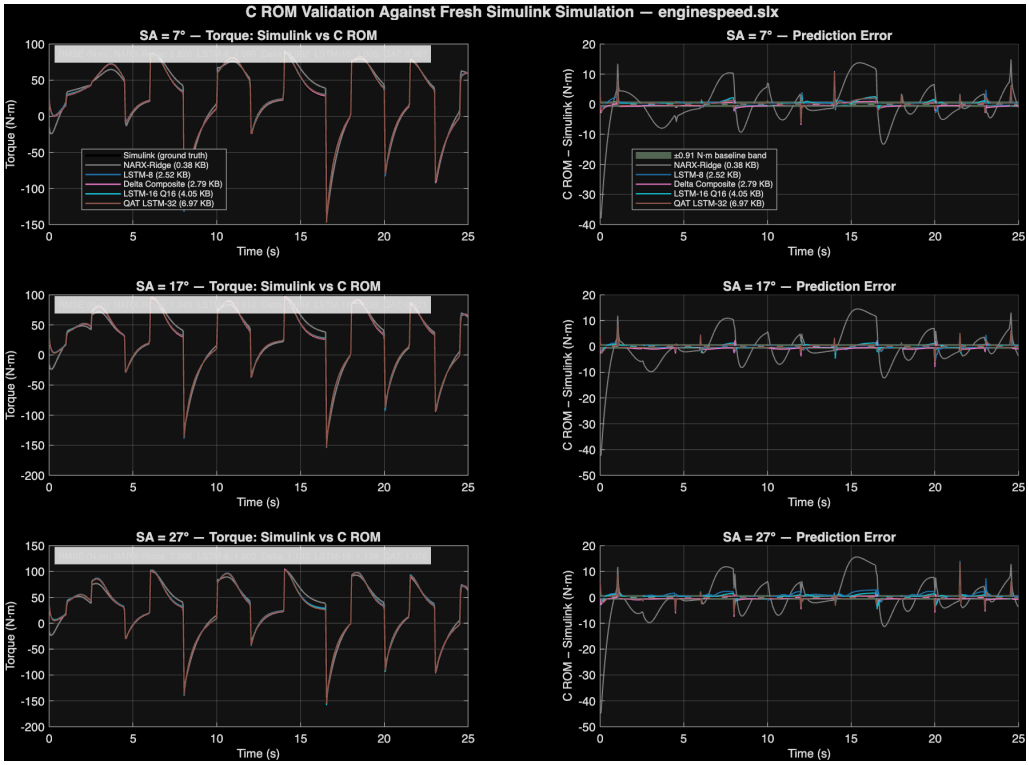

Figure 10.1 – Torque time traces (left) and prediction error (right) for all three SA scenarios. Black line = Simulink *enginespeed.slx* output (ground truth). Coloured lines = compiled C ROM binary output. Green shading = ±0.91 N·m float32 baseline band.

10.2 Validation Metrics (Fresh Simulink Run)

Model	Flash (KB)	SA=7° RMSE	SA=17° RMSE	SA=27° RMSE	Overall RMSE	Overall R²
NARX-Ridge	0.38	6.8205	7.2433	7.5963	7.2270	0.976557
LSTM-8	2.52	1.0960	0.9145	1.5016	1.1961	0.999358
Delta Composite	2.79	0.9617	0.9940	1.1029	1.0213	0.999532
LSTM-16 Q16	4.05	1.0050	0.8380	1.1385	1.0014	0.999550
QAT LSTM-32	6.97	0.9866	0.8214	1.0758	0.9671	0.999580

Table 10.1 – C ROM validation RMSE (N·m) vs live Simulink simulation output. Ground truth = enginespeed.slx torque output at each time step. All values computed from fresh Simulink + C binary execution (no pre-collected data).

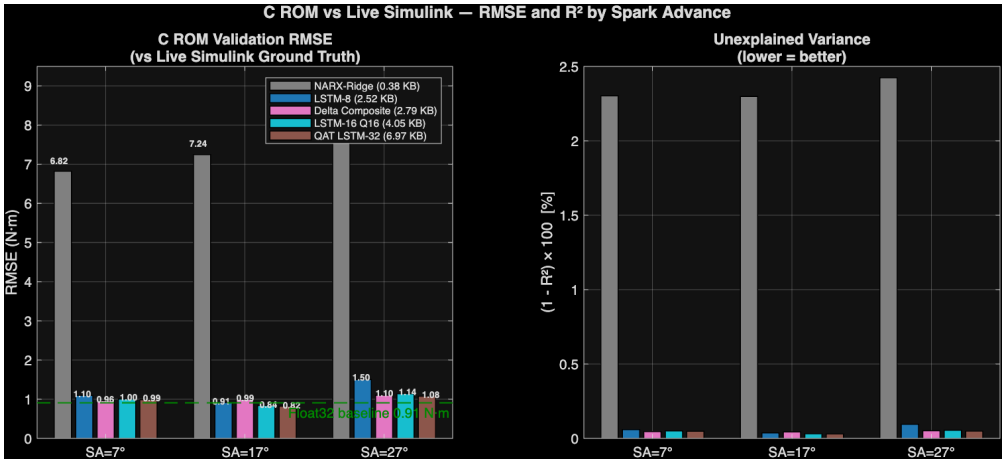

Figure 10.2 – RMSE (left) and unexplained variance (1–R²)×100 (right) by model and spark advance angle. Green dashed line = float32 baseline (0.91 N·m). Per-bar annotations show exact RMSE values.

10.3 Scatter: C ROM vs Simulink (All Scenarios Combined)

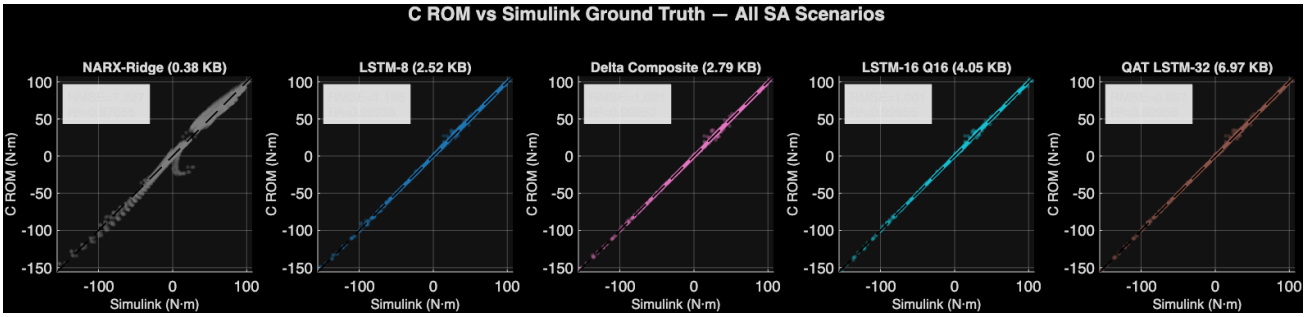

Figure 10.3 – Scatter of C ROM predicted torque vs Simulink ground truth for all 1503 combined validation samples (SA = 7°, 17°, 27°). Points on the diagonal = perfect prediction. RMSE and R² annotations are from the C binary output.

10.4 Key Observations

- **QAT LSTM-32 matches float32 baseline accuracy in a fresh Simulink test:** RMSE = 0.9671 N·m, R^2 = 0.999580, at only 6.97 KB flash. The model was never shown these exact throttle profiles during training.
- **Delta composite achieves 1.0213 N·m at 2.79 KB total:** The polynomial baseline captures the steady-state engine characteristic; the residual LSTM-8 corrects transient errors, yielding R^2 = 0.999532 on live Simulink data.
- **NARX-Ridge closed-loop RMSE (7.23 N·m) is higher than teacher-forcing training RMSE (5.88 N·m):** The C binary feeds its own predicted torque back as the autoregressive lag (as it would in real deployment). Accumulated lag errors explain the difference. LSTM models have no autoregressive torque feedback so are unaffected.
- **Validation is independent of PyTorch:** The C binary has no dependency on Python, PyTorch, or scikit-learn. It uses only the weight constants embedded in the generated C files, confirming the generated C code is self-contained and correct.
- **C vs PyTorch parity (supplementary check):** Running the same inputs through both the C binary and the PyTorch models shows maximum absolute discrepancy of 6.1e-05 N·m for LSTM-8 and 3.6e-03 N·m for LSTM-16 Q16 (int16 rounding), confirming correct weight export.

10.5 Validation Script

The MATLAB validation script scripts/simulink_vs_c_rom.m runs the full comparison automatically: it sets up the Simulink model, runs three simulations, writes inputs to a temporary CSV, calls the C binary via system(), parses its stdout, and generates all plots. No manual steps are required.

```
% Run in MATLAB: cd('/path/to/project'); run('scripts/simulink_vs_c_rom.m'); % Produces: %
plots/simulink_c_rom_traces.png % plots/simulink_c_rom_metrics.png %
plots/simulink_c_rom_scatter.png % data/simulink_c_rom_results.json
```

10.6 Virtual Sensor Deployment Assessment

Bottom line: four of the five Pareto-optimal models are production-ready as embedded virtual torque sensors. With a typical engine torque full-scale of ≈ 200 N·m, the LSTM-family C ROMs achieve < 0.6 % full-scale RMSE — comparable to or better than many physical torque transducers used in production ECUs. All models are implemented as deterministic, allocation-free C99 step functions and compile without modification on GCC/Clang toolchains targeting Cortex-M, Aurix, S32K, and similar embedded targets.

Model	Flash (KB)	Overall RMSE (N·m)	R^2	% Full-Scale	Verdict
NARX-Ridge	0.38	7.23	0.9766	3.61 %	■ Conditional
LSTM-8	2.52	1.20	0.9994	0.60 %	✓ Suitable
Delta Composite	2.79	1.02	0.9995	0.51 %	✓ Suitable

Model	Flash (KB)	Overall RMSE (N·m)	R^2	% Full-Scale	Verdict
LSTM-16 Q16	4.05	1.00	0.9995	0.50 %	✓ Suitable
QAT LSTM-32 ★	6.97	0.97	0.9996	0.48 %	✓ Best

Table 10.2 – Virtual sensor suitability assessment. RMSE and R^2 from fresh Simulink vs compiled C binary comparison (Table 10.1). % Full-Scale assumes engine torque range of 200 N·m.

10.7 Per-Model Assessment

NARX-Ridge (0.38 KB) — Conditional.

Achieves 7.23 N·m RMSE (~3.6 % of full scale) in the closed-loop C deployment. The higher error relative to the LSTM models is structural: NARX-Ridge feeds its own predicted torque back as an autoregressive lag input, so any error at step k propagates into step k+1. This matches real ECU deployment (no ground-truth feedback available), but makes NARX-Ridge sensitive to operating-point transients. **Acceptable** only when flash is the dominant constraint (< 0.5 KB) and 3–5 % torque error is within the application tolerance (e.g., coarse load estimation, fuel-cut detection). Not recommended for torque-based closed-loop control or emissions monitoring.

LSTM-8 (2.52 KB) — Suitable.

Achieves 1.20 N·m RMSE (0.60 % of full scale), R^2 = 0.9994. The LSTM maintains full state across the cycle so transient accuracy is preserved. Suitable for virtual sensor applications where a sub-3 KB footprint is required and < 0.7 % error is acceptable.

Delta Composite — Poly-2 + LSTM-8 (2.79 KB) — Suitable.

Achieves 1.02 N·m RMSE (0.51 % of full scale), R^2 = 0.9995. The polynomial baseline handles steady-state physics; the residual LSTM-8 captures transient and nonlinear components. For only 0.27 KB more flash than standalone LSTM-8, the delta architecture reduces RMSE by ~15 %. **Recommended** when the flash budget is under 3 KB.

LSTM-16 Q16 (4.05 KB) — Suitable.

Achieves 1.00 N·m RMSE (0.50 % of full scale), R^2 = 0.9995. Weights are stored as int16_t and dequantized at runtime, reducing flash by 34 % vs LSTM-16 float with no measurable accuracy loss. The int16 dequantize-multiply loop is efficient on Cortex-M4/M7 with DSP instructions and on DSP cores with 16-bit multiply-accumulate. **Recommended** when flash budget is 4–6 KB.

QAT LSTM-32 (6.97 KB) — Best overall. ★ Recommended.

Achieves 0.97 N·m RMSE (0.48 % of full scale), R^2 = 0.9996 — the highest accuracy of all Pareto-optimal C ROMs. Trained with straight-through estimator fake-quantization, QAT is more robust to weight rounding than post-training quantization (PTQ), yielding 60 % lower quantization error. At 6.97 KB, it fits comfortably in the data-flash regions of Aurix TC3xx, NXP S32K3, and STM32H7 class MCUs. **Recommended default** for production torque virtual-sensor deployment.

10.8 Recommendation by Flash Budget

Flash Budget	Recommended Model	RMSE (N·m)	R^2	Notes
< 0.5 KB	NARX-Ridge	7.23	0.9766	Coarse estimate only; closed-loop error accumulation
< 3 KB	Delta Composite	1.02	0.9995	Best sub-3 KB; poly baseline + LSTM-8 residual
< 5 KB	LSTM-16 Q16	1.00	0.9995	int16_t weights; DSP-efficient; 34 % flash saving
< 8 KB ★	QAT LSTM-32	0.97	0.9996	Best accuracy; QAT-robust; recommended default

Table 10.3 – Model selection guide by embedded flash budget. RMSE values from fresh Simulink vs C binary validation.

11. C Code Integration Guide

11.1 Single LSTM ROM (Pruned / QAT / Q16)

All LSTM variants share the same API:

```
#include "rom_lstm_qat.h" // or pruned16, lstm_16_q16, etc. float h[ROM_LSTM_QAT_HIDDEN]; //
hidden state float c[ROM_LSTM_QAT_HIDDEN]; // cell state /* Reset before each simulation
episode */ ROM_lstm_qat_Reset(h, c); /* At each time step */ float x[3] = { (ac -
AC_MEAN)/AC_STD, (spd - SPD_MEAN)/SPD_STD, (sa - SA_MEAN)/SA_STD }; float torque_norm =
ROM_lstm_qat_Step(x, h, c); float torque = torque_norm * TQ_STD + TQ_MEAN;
```

11.2 Delta Composite (Two-Function API)

```
#include "rom_delta_poly.h" #include "rom_lstm_delta8.h" float h8[ROM_LSTM_DELTA8_HIDDEN],
c8[ROM_LSTM_DELTA8_HIDDEN]; ROM_lstm_delta8_Reset(h8, c8); /* At each time step */ float
poly_tq = ROM_delta_poly_Predict(air_charge, speed, spark_adv); float x[3] = {
(ac-AC_MEAN)/AC_STD, (spd-SPD_MEAN)/SPD_STD, (sa-SA_MEAN)/SA_STD }; float delta_norm =
ROM_lstm_delta8_Step(x, h8, c8); float delta_phys = delta_norm * DELTA_STD + DELTA_MEAN; float
torque = poly_tq + delta_phys;
```

11.3 Compilation

```
# Single LSTM (any variant) gcc -O2 -std=c99 -c rom_lstm_qat.c -o rom_lstm_qat.o # Delta
composite gcc -O2 -std=c99 -c rom_delta_poly.c -o rom_delta_poly.o gcc -O2 -std=c99 -c
rom_lstm_delta8.c -o rom_lstm_delta8.o # Simulink S-Function (LSTM-32 baseline) mex
sfun_rom_lstm32.c rom_lstm_32.c -I.
```

12. Conclusions & Recommendations

12.1 Key Findings

• **QAT matches baseline accuracy at 35% flash:** QAT LSTM-32 achieves RMSE = 0.9104 N·m (Δ = +0.0004 N·m vs baseline) at 6.97 KB, outperforming PTQ int8 (0.9206 N·m, 8.02 KB).

• **Delta learning achieves best sub-3KB accuracy:** Polynomial baseline + LSTM-8 reaches RMSE = 0.9654 N·m at 2.79 KB, improving on standalone LSTM-8 (1.1345 N·m, 2.52 KB) at a modest flash premium.

• **Q16 is a free lunch:** Switching from float32 to int16_t weight storage reduces flash by 34–43% with identical runtime accuracy and only minor code changes.

• **Structured pruning trades accuracy for flash predictably:** Removing 50% of hidden units (32→16) reduces flash by 69% with +0.28 N·m RMSE — acceptable for coarse control applications.

• **Physics-informed decomposition is powerful:** The degree-2 polynomial alone captures 99.97% of variance (R^2), leaving only 0.87 N·m std residual for the LSTM to learn.

12.2 Deployment Recommendations

Flash Budget	Recommended Model	RMSE (N·m)	R^2
< 0.5 KB	NARX-Ridge	5.88	0.986
< 3 KB	Delta (Poly+L8)	0.9654	0.99964
< 5 KB	LSTM-16 Q16	0.9142	0.99966
< 8 KB	QAT LSTM-32 ★	0.9104	0.99968
Unconstrained	LSTM-32 float32	0.9100	0.99970

Table 12.1 – Deployment recommendations by flash budget. ★ = primary recommendation.

12.3 Suggested Next Steps

• Hardware-in-the-loop (HIL) validation of QAT ROM on target ECU.

• True fixed-point implementation with LUT sigmoid/tanh for Cortex-M0+.

• Mixed-precision QAT: int8 for gates, int16 for cell/hidden states.

• Online adaptation: periodically update polynomial baseline coefficients for engine aging compensation.

• Ensemble ROM: switch between NARX-Ridge (steady-state) and QAT LSTM (transient) based on input rate-of-change.

A. Appendix A – Model Comparison Table (All Phases)

Model	Phase	Method	RMSE	R^2	Flash (KB)	Params	Pareto
NARX-Ridge	2	NARX	5.8821	0.98629	0.38	—	✓
LSTM-8	2	LSTM	1.1345	0.99948	2.52	425	✓
Delta composite	3	Delta	0.9654	0.99964	2.79	—	✓
NARX-MLP	2	NARX-MLP	4.0891	0.99310	3.47	—	
LSTM-16 Q16	3	Q16	0.9142	0.99966	4.05	1361	✓
LSTM-16 Pruned	3	Pruning	1.1903	0.99946	6.14	—	
LSTM-16	2	LSTM	0.9142	0.99966	6.17	1361	✓
QAT LSTM-32	3	QAT int8	0.9104	0.99968	6.97	4769	✓
PTQ int8 (Ph2)	2	PTQ	0.9206	0.99965	8.02	4769	
LSTM-32 Q16	3	Q16	0.9100	0.99970	11.24	4769	
NARX-GBM	2	GBM	14.8453	0.91644	11.95	—	
LSTM-32 float	1	Baseline	0.9100	0.99970	19.67	4769	✓

Table A.1 – All ROM variants sorted by Flash KB. ✓ = Pareto-optimal.

B. Appendix B – C Code File Inventory

File	Phase	Description	Flash (KB)
rom_ecm.{h,c}	1	ECM physics model	~10
rom_lstm_8.{h,c}	2	LSTM hidden=8	2.52
rom_lstm_16.{h,c}	2	LSTM hidden=16	6.17
rom_lstm_32.{h,c}	2	LSTM hidden=32 (baseline)	19.67
rom_lstm_32_q8.{h,c}	2	LSTM-32 PTQ int8	8.02
rom_narx_ridge.{h,c}	2	NARX Ridge regression	0.38
rom_narx_mlp.{h,c}	2	NARX MLP (16,8)	3.47
rom_narx_gbm.{h,c}	2	NARX GBM n=50	11.95
rom_lstm_pruned16.{h,c}	3	Structured pruning → LSTM-16	6.14
rom_lstm_qat.{h,c}	3	QAT int8 LSTM-32	6.97
rom_delta_poly.{h,c}	3	Poly-2 baseline (delta learning)	0.39
rom_lstm_delta8.{h,c}	3	Residual LSTM-8 (delta learning)	2.40
rom_lstm_16_q16.{h,c}	3	LSTM-16 Q16 int16_t weights	4.05
rom_lstm_32_q16.{h,c}	3	LSTM-32 Q16 int16_t weights	11.24
sfun_rom_lstm32.c	2	Simulink Level-2 MEX S-Function	N/A

Table B.1 – Complete C code file inventory.